

DSC 335, Handout 04: The Beta-Binomial Bayesian Model (*Chapter 3*)

LEARNING OBJECTIVES.

- *Determine the posterior probability distribution in the case of conjugate priors.*

Motivating Example

What is the probability that if you spin a penny on its side, it will land on heads? It's typically not 50%. Why?

However, each penny is slightly different because of the age, the amount of dirt on either side, etc. Thus if we randomly choose a penny from circulation, the probability of spinning a heads is unknown to us, and we will treat it as a random variable. Is this a discrete or continuous random variable? What possible values could this random variable take on?

Spin the penny 20 times and record the proportion of times it landed on heads. Below, sketch a graph of the set of class proportions.

Continuous Probability Models

Recall that our parameter of interest is a probability π . Would it make sense to model it with any of the distributions from MAT 130 (Normal, t, Chi-squared)?

The Beta Distribution (Model)

In R: The `plot_beta(alpha,beta)` function in the `bayesrules` package sketches the pdf for a $\text{Beta}(\alpha, \beta)$ distribution.

Example 1. Plot the Beta pdf with various hyperparameters. How does the shape, center, and spread of the distribution change with the hyperparameters?

Expected Value, Mode, and Variance of the Beta Distribution

In R: The `summarize_beta(alpha,beta)` function in the `bayesrules` package gives the mean, mode, variance, and standard deviation for a $\text{Beta}(\alpha, \beta)$ distribution.

So it makes sense to use the Beta model for our prior information about π . But how can we determine which hyperparameters to use? (This is called *tuning* the prior.)

Solve the system of equations based on the mean and variance of the class proportions to determine a reasonable prior for our data. Graph that prior and see how well it matches the graph of the class proportions.

Now that we have our prior, we need a likelihood based on our data. Before class, Dr. Daryl spun a penny 40 times and recorded the number of heads, Y . Conditional on the true probability of heads, π , what distribution can we use to model Y ? What is the pdf of that distribution?

It turns out that $Y = 17$ of Dr. Daryl's spins were heads. What is the likelihood function, $L(\pi|Y = 17)$? In plain English, what does this function represent?

Now that we have the prior and the likelihood, we want to combine them into the posterior.

In R: The `plot_beta_binomial(alpha, beta, y, n)` function in the `bayesrules` package in R plots by default the prior, scaled likelihood, and posterior of a beta-binomial model. (You can hide any of those three functions, for example by using the option `posterior=FALSE`).

Use the function above to plot the prior, likelihood, and posterior for our example. What do you notice?

In R: The `summarize_beta_binomial(alpha, beta, y, n)` gives the parameters, mean, mode, variance, and standard deviation of the prior and posterior Beta distributions in the beta-binomial model.

What is the posterior distribution for our example? Write a couple of sentences explaining what that says about the question of interest?

Deriving the Beta-Binomial Posterior

Conjugate Priors

Example 2. For this question assume that the class data survey is representative of all Centre students. Suppose we are interested in π , the true proportion of Centre students for whom TikTok is their preferred social media. Using a web search or generative AI, develop a reasonable prior for π based perhaps on nationwide trends or your own intuition. Then use the class survey data to construct a likelihood. Finally, combine those into a posterior distribution.

Example 3. (a) A computer game randomly determines its difficulty level for all games played that day. You are on a group chat with some friends who have played it that day, and the consensus is that the probability of beating the game is low, maybe around 30%, although some think it is as low as 15% and others that it is as high as 50%. Give a reasonable prior for these beliefs.

(b) Suppose you play the game 20 times and win 9 of them. Derive the posterior distribution of the probability of winning a game given the data.

(c) If instead you had played 200 times and won 90, what affect would that have on the posterior. Make a guess, and then calculate the actual posterior to see if you are correct.